1 **(a)** 2 hours 15 minutes **[B1]**

(b) Adds 2 hours 35 minutes to 16 45 (e.g., 18 80) **[M1]**

19 20 **[A1]**

[Total Marks: 3]

2 Arrival time in London local time = 21 : 45 + 6 hours 50 minutes **[M1]**

04 : 35 (next day) **[M1]**

Dubai local time = 04 : 35 + 4 hours = 08 : 35 **[A1]**

[Total Marks: 3]

3 **(a)** London departure time = 10 30 or Dubai arrival time = 23 15 **[M1]**

8 hours 45 minutes **[A1]**

(b) London departure time = 18 45 or Dubai arrival time = 06 05 (Saturday) **[M1]**

02 05 Saturday **[A1]**

[Total Marks: 4]

4 Flight arrival time in Dubai local time: 14 30 + 14 h 15 min = 28 45 (or 04 45 next day) **[M1]**

Adds time zone difference: 28 45 + 7 hours (or 04 45 + 7 hours) **[M1]**

11 45 **[A1]**

[Total Marks: 3]

5 **(a)** $3 \times 10^5 \times 60 \times 60 \times 24 \times 365$ **[M1]**

9.46×10^{12} or 9.4608×10^{12} **[A1]**

(b) $\dfrac{4.5 \times 10^{13}}{9.4608 \times 10^{12}}$ **[M1]**

4.76 **[A1]**

[Total Marks: 4]

6 **(a)** 11 : 03 − 08 : 15 **[M1]**

2 hours 48 minutes **[A1]**

(b) 48 minutes $= \dfrac{48}{60}$ hours = 0.8 hours **[M1]**

Average speed $= \dfrac{420}{2.8}$ **[M1]**

150 **[A1]**

[Total Marks: 5]

7 **(a)** $920 \div 24$ **[M1]**

38.333... hours **[A1]**

38 hours 20 minutes **[A1]**

(b) Adds their time from part (a) to Wednesday 17 50 **[M1]**

Wednesday 17 50 + 20 minutes = 18 10 **[M1]**

Wednesday 18 10 + 38 hours = Thursday 18 10 + 14 hours **[M1]**

Day: Friday **[A1]**

Time: 08 10 **[A1]**

[Total Marks: 7]

8 **(a)** $\frac{6600}{800}$ or 8.25 **[M1]**

8 hours 15 minutes **[A1]**

(b) Arrival time in City A time = 05 55 **[M1]**

00 55 Wednesday **[A1]**

(c) Departure from City B = 15 15 (local time) or 20 15 (City A time) **[M1]**

Arrival in City B time = 23 00 **[M1]**

04 00 Thursday **[A1]**

[Total Marks: 7]

9 **(a)** Minutes in a year = $365 \times 24 \times 60$ **[M1]**

525600 minutes **[M1]**

Number of orbits = $\frac{525600}{92} = 5713.043\ldots$ **[M1]**

5713 **[A1]**

(b) **(i)** $\frac{11440}{880} = 13$ **[B1]**

(ii) Time difference = $9 - (-4) = 13$ hours (Q is 13 hours ahead) **[M1]**

Departure time in City Q's local time = 22 : 45 (Monday) +13 hours = 11 : 45 (Tuesday) **[M1]**

Arrival time in City Q's local time = 11 : 45 (Tuesday) +13 hours (flight time) = 00 : 45 (Wednesday) **[A1]**

[Total Marks: 8]

10 **(a)** 115 minutes = 1 hour 55 minutes **[M1]**

21 30 (or 21 : 30) **[A1]**

(b) Arrival time in New York $= 21\,40 + 7$ h 15 m $= 04\,55$ (next day) **[M1]**

Arrival time in Paris $= 04\,55 + 6$ hours **[M1]**

$10\,55$ (or $10:55$) **[A1]**

[Total Marks: 5]

11 (a) Time $= \dfrac{42.2}{12.5}$ or 3.376 hours **[M1]**

$0.376 \times 60 = 22.56$ minutes **[M1]**

$0.56 \times 60 = 33.6$ seconds **[M1]**

3 hours, 22 minutes, 34 seconds **[A1]**

(b) Correct difference calculation, e.g. 12 hours 64 minutes $-$08 hours 35 minutes **[M1]**

4 hours 29 minutes **[A1]**

[Total Marks: 6]

12 (a) $\dfrac{35}{x} + \dfrac{42}{x-5}$ **[B1]**

(b) 4 hours 33 minutes $= 4 + \dfrac{33}{60}$ or $\dfrac{91}{20}$ hours **[B1]**

$\dfrac{35}{x} + \dfrac{42}{x-5} = \dfrac{91}{20}$ **[M1]**

Multiplies by common denominator: $20 \times 35(x-5) + 20 \times 42x = 91x(x-5)$ **[M1]**

$700x - 3500 + 840x = 91x^2 - 455x$ **[M1]**

$1540x - 3500 = 91x^2 - 455x \implies 91x^2 - 1995x + 3500 = 0$ **[M1]**

Divides by 7 to reach $13x^2 - 285x + 500 = 0$ without errors **[A1]**

(c) Uses quadratic formula or factors: $(13x - 25)(x - 20) = 0$ **[M1]**

$x = 20$ or $x = \dfrac{25}{13}$ (or 1.92...) **[A1]**

$x = 20$ only (rejects $\dfrac{25}{13}$ as speed $x - 5$ must be positive) **[A1]**

[Total Marks: 8]